

David Antwi

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EDUCATION

Yale University

B.S. Electrical Engineering and Computer Science

New Haven, CT

Expected May 2028

- **Coursework:** Data Structures, Systems Programming, Software Engineering, Full-Stack Web, Intro to Machine Learning, Intro to AI, HCI, Linear Algebra, Discrete Math

EXPERIENCE

BENMP

Software Engineer, Tech Team (AI and Full-Stack)

Jul 2026 – Present

International nonprofit, 30+ countries

- Built and launched the organization's partner management platform, now holding 16,000+ partners across 30+ countries and used by admins at 97 regional hubs.
- Built payment reconciliation for bank and mobile money, plus guardrailed, provider-agnostic WhatsApp automation.
- Built a grounded, read-only AI assistant over partner data: code computes every figure, the model only explains it.
- Migrated 26,000+ records with zero loss and scaled from 1 to 30+ countries in a month via a data-driven region model.

Beakr, Inc. (Stealth Startup)

Software Engineering Intern, AI memory platform for life sciences research

May 2026 – Aug 2026

New York, NY

- Designed and shipped Beakr Interviews, an AI system that finds gaps in a lab's knowledge base, interviews the right researcher by text or voice, and turns their answers into cited wiki entries.
- Led the computer-use desktop agent: rebuilt search over 80K+ files, made every answer cite its source file, and got it running reliably on Windows by fixing 35 release-blocking defects (Rust).
- Extended the desktop agent to launch and supervise coding agents (Claude Code, Codex) on the user's computer, with explicit approval before each run and a full audit log.
- Built five evaluation datasets on agent reliability, memory, and tool use, published to Hugging Face.

SynovAI

Software Engineering Intern, AI retrosynthesis startup

Oct 2025 – May 2026

New Haven, CT

- Designed and built a vendor-aware stock filtering feature that shows researchers only routes whose starting materials are purchasable, across 5 vendor catalogs of 2.5M+ compounds.
- Built a chemical-safety layer that pulls GHS hazard data from PubChem and flags hazards on each molecule in a route.
- Built a self-serve fine-tuning pipeline (PyTorch, QLoRA) covering validation, training, and automated evaluation.
- Led a UI/UX overhaul of the researcher platform (React, Next.js), redesigning the settings and results views.

Yale Model African Union

Lead Developer

Nov 2025 – Present

New Haven, CT

- Replaced the Squarespace site with a custom Next.js rebuild of yalemodelau.org, plus a registration platform with Stripe payments that has taken 1,500+ applications.
- Built the board's review desk for delegate applications and financial aid, with blind review and role-based access.
- Built the previous conference's delegate training platform, serving 335+ users.

PROJECTS

Intentionality | AI-Powered Gamified Productivity Platform

2026

- Launched a gamified productivity app with streaks and friend accountability, used by 300+ people.
- Built an AI planner grounded in users' calendars and coursework, with failover across two LLMs.

Better Office Hours | 1st Place, YaleAI x SpaceXAI Cursor Build Challenge

2026

- Built a real-time multimodal voice agent tutor that talks students through concepts and problems while drawing diagrams and equations on a live whiteboard or highlighting their uploaded notes.
- Built a custom SVG whiteboard engine with local LaTeX rendering and keyframe animations timed to the agent's speech.
- Built the voice loop: on-device speech detection, ElevenLabs transcription and speech, and interruption-safe turn-taking.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C++, C, Rust

Frameworks & Libraries: React, Next.js, React Native, FastAPI, Flask, Spring Boot, PyTorch, Anthropic SDK

Tools: PostgreSQL, Supabase, Redis, Docker, Git, CI/CD, Linux, Hugging Face, Claude Code, Cursor

HONORS

Ramp Builders Cup (2026): Best Use of Cursor for TicketDodge, which predicts parking ticket odds across New York.

Google x Yale SOM Build with AI Hackathon (2026): 2nd place for Ocura, glasses for the visually impaired.

Prometheus X Catapult Fellow: \$5,000 grant. **YC Startup School 2026:** selected from 30,000+ applicants.